

Skills Summary

Vector Motion, Polar Intersections, and Accumulated Quantities

Use this reference while completing your worksheet or when studying.

Skill 1 — Motion given as a vector

From the velocity $\langle x'(t), y'(t) \rangle$:

- speed = $\sqrt{(x')^2 + (y')^2}$ — a length, so the components go under one root
- position = starting point + the integral of each component separately
- distance travelled = $\int_a^b \text{speed } dt$; displacement compares endpoints only

Example: A particle has $x'(t) = t$ and $y'(t) = 2$. Find its speed at $t = 3$.

Step 1. Speed is the length of the velocity vector, so evaluate both components at that instant and take the magnitude.

Step 2. At $t = 3$ the velocity is $\langle 3, 2 \rangle$, so the speed is $\sqrt{9 + 4} = \sqrt{13}$.

$\Rightarrow$ **3.606**

Try it: A particle has $x'(t) = t$ and $y'(t) = 5$. Find its speed at $t = 2$.

check: 5.385

Skill 2 — Where two polar curves meet

Two steps, always both. (1) Set $r_1(\theta) = r_2(\theta)$ and solve on $[0, 2\pi)$. (2) Check the pole separately — each curve may reach $r = 0$ at a different θ , so the algebra in step 1 cannot see it. For the area between two curves on an interval where one is outside the other, $A = \frac{1}{2} \int (r_{\text{out}}^2 - r_{\text{in}}^2) d\theta$ between consecutive intersection angles.

Example: Find every θ in $[0, 2\pi)$ where $r = 2$ meets $r = 4 \sin \theta$.

Step 1. Both curves are given as r in terms of θ , so equate the two expressions and solve the resulting trigonometric equation over one full turn.

Step 2. $4 \sin \theta = 2$ gives $\sin \theta = \frac{1}{2}$, whose solutions on $[0, 2\pi)$ are the two first-and-second-quadrant angles.

$\Rightarrow$ $\theta = \frac{\pi}{6}, \frac{5\pi}{6}$

Try it: Find every θ in $[0, 2\pi)$ where $r = 3$ meets $r = 6 \cos \theta$.

check: $\frac{\pi}{3}, \frac{2\pi}{3} = \theta$

Skill 3 — Accumulating a rate

Rule: the total accumulated between $t = a$ and $t = b$ is $\int_a^b r(t) dt$ — integrate the rate, do not divide by anything. Check the units: litres per minute times minutes gives litres. If the question instead asks *when* the rate reaches some level, that is an equation to solve, not an integral.

Example: Water enters at $r(t) = 4 + 6t$ litres per minute. How much enters in the first 3 minutes?

Step 1. "How much enters" asks for a total, so integrate the rate across the given time interval.

Step 2. $\int_0^3 (4 + 6t) dt = [4t + 3t^2]_0^3 = 12 + 27.$
 $\Rightarrow$ **39** litres

Try it: Water enters at $r(t) = 5 + 4t$ litres per minute. How much enters in the first 4 minutes?

check: **52** litres